

RHCSA_v8.2

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***Important_Instructions: Please read carefully.**

hostname: servera.lab.example.com (172.25.250.10)

hostname: serverb.lab.example.com (172.25.250.11)

- You will be given by 2 VMs
 - Total number of Questions will be around 22
 - In one system root password is already set (no need to reset) but in second system password need to be recovered.
 - In your both system root passwd is "trootent"
 - In one system Network configuration is required but in another one networking is already done
 - NTP need to be configured in only one system (not in both)
 - YUM Repo need to configured in both systems.
 - There is not any Q to configure LDAP Client (it is already configured).
 - You just need to configure automounting for LDAP user's Home DIR in one system. (follow same steps as RHEL-7)
 - Firewall and SELinux both will be pre-enabled.
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Server-a:

#Q1. Configure network and set the static hostname.

IP ADDRESS	= 172.25.250.10
NETMASK	= 255.255.255.0
GATEWAY	= 172.25.250.254
DNS	= 172.25.250.254
Domain name	= lab.example.com
hostname	= servera.lab.example.com

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#Q2. Configure YUM repos with the given link (2 repos: 1st is Base and 2nd is AppStream)

- Base_url= http://content.example.com/rhel8.0/x86_64/dvd/BaseOS
 - AppStream_url= http://content.example.com/rhel8.0/x86_64/dvd/AppStream
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#Q3. Debug SELinux:

- A web server running on non standard port 82 is having issues serving content. Debug and fix the issues.
- The web server on your system can server all the existing HTML files from /var/www/html (NOTE: Do not make any changes to these files)
- Web service should automatically start at boot time.

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#Q4. Create User accounts with supplementary group.

- Create the group a named "**sysadms**".
- Create users as named "**natasha**" and "**harry**", will be the supplementary group "**sysadms**".
- Create a user as named "**sarah**", should have non-interactive shell and it should be not the member of "**sysadms**".
- Password for all users should be "**trootent**"

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#Q5. Configure a cron job that runs every 1 minutes and executes:

logger "EX200 in progress" as the user **natasha**.

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#Q6. Create a collaborative Directory.

- Create the Directory **"/home/manager"** with the following characteristics.
- Group ownership of **"/home/manager"** should go to "**sysadms**" group.
- The directory should have full permission for all members of "**sysadms**" group but not to the other users except "**root**".
- Files created in future under **"/home/manager"** should get the same group ownership .

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#Q7. Configure NTP

- Synchronize time of your system with the server **classroom.example.com**.

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#Q8. Configure AutoFS

- All **Ldapuser2** home directory is exported via **NFS**, which is available on **classroom.example.com (172.25.254.254)** and your NFS-exports directory is **/home/guests** for **Ldapuser2**,
- **Ldapuser2's** home directory is **classroom.example.com:/home/guests/ldapuse2**
- **Ldapuser2's** home directory should be automount autofs service.
- Home directories must be **writable** by their users.
- while you are able to log in as any of the user **ldapuser1** through **ldapuser20**, the only home directory that is accessible from your system is **ldapsuser2**

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#Q9. ACL.

- Copy the file /etc/fstab to /var/tmp/ and configure the "ACL" as mentioned following.
 - The file /var/tmp/fstab should be owned by the "root".
 - The file /var/tmp/fstab should belong to the group "root".
 - The file /var/tmp/fstab should not be executable by any one.
 - The user "sarah" should be able to read and write to the file.
 - The user "harry" can neither read nor write to the file.
 - Other users (future and current) should be able to read /var/tmp/fstab.
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#Q10. Create user 'bob' with 2112 uid and set the password 'trootent'

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#Q11. Locate all files owned by user "harry" and copy it under /root/harry-files

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#Q12. Find a string 'ich' from "/usr/share/dict/words" and put it into /root/lines file.

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#Q13. create an archive '/root/backup.tar.bz2' of /usr/local directory and compress it with bzip2.

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Server-2:

NOTE: In this Server 3 Disks will be given.

1. /dev/vda : for ROOT filesystem (don't do anything under this Disk)
 2. /dev/vdb : You need to use Swap and LVM Partition.
 3. /dev/vdc : Will be used for Stratis.
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#Q14. Reset root user password and make it 'trootent'

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#Q15. Configure YUM Repos

- Base_url= "http://content.example.com/rhel8.0/x86_64/dvd/BaseOS"
- AppStream_url= "http://content.example.com/rhel8.0/x86_64/dvd/AppStream"

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#Q16. Resize a logical Volume

- Resize the logical volume **"mylv"** so that after reboot the size should be in between **200MB** to **300MB**.
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#Q17. Add a swap partition of 512MB and mount it permanently.

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#Q18. Create a logical Volume and mount it permanently.

- Create the logical volume with the name **"wshare"** by using **50PE's** from the volume group **"wgroup"**.
 - Consider each **PE size** of the volume group as **"8 MB"**.
 - Mount it on **/mnt/wshare** with file system **vfat**.
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#Q19. Create a new STRATIS volume according to following requirements:

- Use the unpartitioned disk
- The volume is named **'stratisfs'** belongs to **'stratispool'**
- The volume must be mounted permanent under **'/stratisvolume'**
- Place a copy of the file **"http://classroom.example.com/content/Rhcsa-v8/rhel-8_Ex200_Q_1"** under **'/stratisvolume'**
- Take a **snapshot** of **stratisfs** named **stratissnap**.

(OR)

Create a new VDO partition using to following requirements:

- Use the unpartitioned disk
 - Vdo name **"Vdo1"** and logical size should be **50GB**
 - Mount it on **/vdomount** permanently with file system **xfs**.
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#Q20. Configure System Tuning:

- Choose the recommended **'tuned'** profile for your system and set it as the default.

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#Q21.

- Create a container **logserver** from an image **rsyslog** in node1 From **registry.lab.example.com**
 - Configure the container with systemd services by an existing user "**Walhalla**".
 - Service name should be **container-logserver**, and configure it to start automatically across reboot.
- Node1: root steps

```
# useradd user1
# passwd user1
# yum module install container* -y
# ll /var/log/
# vim /etc/systemd/journald.conf
[Journal]
Storage=persistent
:wq!
/run/log
# systemctl restart systemd-journald
# ll /run/log
# ll /var/log/
# su - user1
# mkdir /home/user1/container-logserver/
# exit
# cp -r /var/log/journal/ /home/user1/container-logserver/
# chown -R user1:user1 /home/user1/container-logserver/
# systemctl restart systemd-journald
# reboot
after bootup do ssh to user1 and continue:
# ssh user1@servera.lab.example.com
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#Q22

- Configure your host journal to store all journal across reboot
 - Copy all *.journal from /var/log/journal and all subdirectories to **/home/Walhalla/container_logserver**
 - Configure automount **/var/log/journal** from **logserver** (container) to **/home/walhalla/container_logserver** when container starts.
- node1:user1 steps

```
# podman login registry.redhat.io
# username:
# password:
# podman search rsyslog
# podman pull registry.redhat.io/rhel8/rsyslog
# podman image list
# podman run -d --name logserver -v /home/user1/container-logserver:/var/log/journal:Z
registry.redhat.io/rhel8/rsyslog
# podman container list
# podman ps
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```
# mkdir -p ~/.config/systemd/user
# cd .config/systemd/user/
# loginctl enable-linger
# loginctl show-user user1
# podman generate systemd --name logserver -f -n
# systemctl --user daemon-reload
'in this step, we went to host and one time we rebooted it'
# systemctl --user enable --now container-logserver.service
# systemctl --user start --now container-logserver.service
# systemctl --user status --now container-logserver.service
# podman exec -it logserver /bin/bash
# ls /var/log/
# exit
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